

# ELEC2134 Week 1: Major Corrections and Worked Solutions

## Topic 1: Periodic Signals and Fourier Series, T3 2026

### Purpose and scope

This focused note includes only corrections that change the mathematical meaning or a stated answer, along with detailed working for examples and tutorial questions where the lecture slides show a compressed or direct answer. Only the corrections explicitly listed below supersede the corresponding slide content. All other slide material remains unchanged.

### 1. Major corrections affecting mathematics or stated answers

- **Slide 11 - rectangular waveform:** the repetition frequency is  $f = 1/T$ , not  $1/\tau$ . Here  $T$  is the period,  $\tau$  is the pulse width, and the duty cycle is  $D = \tau/T$ . A rectangular waveform does not require  $\tau < T/2$ .
- **Slide 20 - full-wave rectifier:** the harmonic terms must be written as  $\cos(2m\omega t)$ . For a 50 Hz input, the full-wave-rectified output has DC plus components at 100, 200, 300, ... Hz; there is no 50 Hz component.
- **Slides 25 and 27 - Fourier coefficient equations:** use the corrected coefficient formulas given in Sections 2.4 and 2.5 below. The equations rendered on the slides are incomplete in places.
- **Slides 29 and 32 - odd symmetry:** odd symmetry gives  $A_0 = 0$  and  $A_n = 0$ , so only sine terms remain. Odd symmetry alone does not remove even-numbered harmonics. Even harmonics vanish when half-wave symmetry is also present.
- **Slide 65, Question 1:** the square-wave answer must contain odd harmonics only. The printed summation over every positive integer  $n$  is incorrect.
- **Slide 65, Question 2:** the compact answer shown on the slide is valid only when the summation is restricted to odd  $n$ .
- **Slide 68, Question 6:** the result is the complex Fourier coefficient  $C_n$ ; the answer box should not be labelled  $v(t)$ .
- **Slide 71, Question 1:** the general  $A_n$  and  $B_n$  formulas must integrate over one complete period. The interval 0 to  $T/2$  is not a valid general formula unless you first state an appropriate symmetry and change the prefactor.

## 2. Detailed worked solutions for lecture examples with direct answers

### 2.1 Slide 11 - rectangular waveform.

The current slide contains the pulse-width/frequency relationship in a form that can be read as

$$f = \frac{1}{\tau}.$$

That is not correct in general. Frequency is determined by the **period**  $T$ :

$$\boxed{f = \frac{1}{T}},$$

while

$$\tau = \text{pulse width}, D = \frac{\tau}{T}$$

is the duty cycle. The current slide also describes a rectangular waveform as requiring its high duration to be shorter than  $T/2$ ; that restriction is unnecessary. A rectangular pulse train can have a duty cycle below, equal to, or above 50%. The slide deck explicitly introduces pulse width and duty cycle in this section.

### 2.2 Slide 15 - average-value questions

For the four small waveforms on the left of the slide:

1. **Sinusoid symmetric about zero:** positive and negative areas cancel, so  $V_{\text{avg}} = 0$ .
2. **Unipolar triangular wave from 0 to  $A$  and back to 0:** area is  $AT/2$ , hence  $V_{\text{avg}} = A/2$ .
3. **Bipolar rectangular wave with equal  $+A$  and  $-A$  durations:**  $V_{\text{avg}} = 0$ .
4. **Unit pulse high for half of each period:**  $V_{\text{avg}} = 1 \times (1/2) = 1/2$ .

#### Half-wave-rectified sine

Let

$$v(t) = \begin{cases} V_m \sin(\omega t), & 0 \leq \omega t \leq \pi, \\ 0, & \pi < \omega t < 2\pi. \end{cases}$$

Since  $T = 2\pi/\omega$ ,

$$V_{\text{avg}} = \frac{1}{T} \int_0^T v(t) dt.$$

$$V_{\text{avg}} = \frac{\omega}{2\pi} \int_0^{\pi/\omega} V_m \sin(\omega t) dt.$$

$$V_{\text{avg}} = \frac{\omega V_m}{2\pi} \left[ -\frac{\cos(\omega t)}{\omega} \right]_0^{\pi/\omega} = \frac{V_m}{\pi}.$$

**Note on the horizontal axis:** The waveform is plotted against phase  $\omega t$ . The calculation above integrates with respect to time  $t$ , hence the phase limits  $\pi$  and  $2\pi$  correspond to the time limits  $\pi/\omega$  and  $2\pi/\omega$ . Equivalently, defining  $\theta = \omega t$  gives the same result by integrating with respect to  $\theta$ .

#### Full-wave-rectified sine

The full-wave waveform repeats every  $\pi$  in the phase variable, so

$$V_{\text{avg}} = \frac{1}{\pi} \int_0^{\pi} V_m \sin \theta \, d\theta = \frac{2V_m}{\pi}.$$

The full-wave average is twice the half-wave average because both half-cycles contribute positively.

## 2.3 Slide 17 - reading a line spectrum

For

$$v(t) = \frac{4}{\pi} \sin(\omega_0 t) + \frac{4}{3\pi} \sin(3\omega_0 t) + \frac{4}{5\pi} \sin(5\omega_0 t),$$

- there is no DC term;
- spectral lines occur at  $\omega_0$ ,  $3\omega_0$  and  $5\omega_0$ ;
- the line heights are  $4/\pi$ ,  $4/(3\pi)$  and  $4/(5\pi)$ ;
- there are no lines at  $2\omega_0$  or  $4\omega_0$ .

This is a **one-sided amplitude line spectrum**. Phase is not shown on this graph.

## 2.4 Slide 20 - half-wave and full-wave rectifier spectra

For an input  $v_s(t) = V_m \sin(\omega_0 t)$ , the half-wave-rectified output is

$$v_{\text{HW}}(t) = \frac{V_m}{\pi} + \frac{V_m}{2} \sin(\omega_0 t) - \frac{2V_m}{\pi} \sum_{m=1}^{\infty} \frac{\cos(2m\omega_0 t)}{4m^2 - 1}$$

It contains:

- DC;
- the input-frequency term at  $f_0$ ;
- even multiples  $2f_0, 4f_0, 6f_0, \dots$

The full-wave-rectified output is

$$v_{\text{FW}}(t) = \frac{2V_m}{\pi} - \frac{4V_m}{\pi} \sum_{m=1}^{\infty} \frac{\cos(2m\omega_0 t)}{4m^2 - 1}$$

It contains DC and components at  $2f_0, 4f_0, 6f_0, \dots$ . For a 50 Hz input, its fundamental ripple frequency is 100 Hz.

## 2.5 Slide 25 - corrected general coefficient formulas

$$A_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) \, dt$$

$$A_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cos(n\omega_0 t) \, dt, \quad n \geq 1$$

$$B_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin(n\omega_0 t) \, dt, \quad n \geq 1$$

The extra factor of 2 in  $A_n$  and  $B_n$  occurs because

$$\int_T \cos^2(n\omega_0 t) \, dt = \int_T \sin^2(n\omega_0 t) \, dt = \frac{T}{2},$$

whereas

$$\int_T 1^2 dt = T.$$

## 2.6 Slide 27 - even symmetry

If the origin is chosen at the centre of even symmetry,  $x(t) = x(-t)$ , then

$$A_0 = \frac{2}{T} \int_0^{T/2} x(t) dt,$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt, \quad B_n = 0.$$

## 2.7 Slides 29-32 - odd symmetry and the odd square wave

For a general odd waveform,

$$x(t) = -x(-t) \Rightarrow A_0 = 0, \quad A_n = 0.$$

Therefore only sine terms remain. Even-numbered sine harmonics may still occur unless the signal also has half-wave symmetry.

For the displayed square wave,

$$x(t) = \begin{cases} -1, & -1 < t < 0, \\ +1, & 0 < t < 1, \end{cases} \quad T = 2,$$

$$\omega_0 = \frac{2\pi}{T} = \pi.$$

Odd symmetry gives  $A_0 = A_n = 0$ . Over  $0 < t < 1$ ,  $x(t) = 1$ , so

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt.$$

$$B_n = 2 \int_0^1 \sin(n\pi t) dt = \frac{2[1 - \cos(n\pi)]}{n\pi}.$$

Thus

$$B_n = \begin{cases} \frac{4}{n\pi}, & n \text{ odd}, \\ 0, & n \text{ even}, \end{cases}$$

and

$$x(t) = \frac{4}{\pi} \left[ \sin(\pi t) + \frac{1}{3} \sin(3\pi t) + \frac{1}{5} \sin(5\pi t) + \dots \right].$$

The even harmonics vanish because the waveform has **half-wave symmetry in addition to odd symmetry**.

## 2.8 Slides 35-36 - Fourier series of $f(x) = x^2$ on $-\pi < x < \pi$

The periodic extension has  $T = 2\pi$  and fundamental angular frequency 1 in the variable  $x$ . Since  $x^2$  is even,  $B_n = 0$ .

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{\pi^2}{3}.$$

For  $n \geq 1$ ,

$$A_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos(nx) dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx.$$

Using integration by parts twice,

$$\int x^2 \cos(nx) dx = \frac{x^2 \sin(nx)}{n} + \frac{2x \cos(nx)}{n^2} - \frac{2 \sin(nx)}{n^3}.$$

Therefore

$$A_n = \frac{4(-1)^n}{n^2},$$

and

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx).$$

The one-sided magnitude spectrum has magnitudes  $4/n^2$ . The coefficient sign alternates, corresponding to phase 0 or  $\pi$  in amplitude-phase form.

## 2.9 Slides 38-39 - unipolar rectangular waveform

Let

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$$

periodically extended with  $T = 2\pi$ . It is neither even nor odd about the selected origin.

$$A_0 = \frac{1}{2\pi} \int_0^{\pi} 1 dx = \frac{1}{2}.$$

$$A_n = \frac{1}{\pi} \int_0^{\pi} \cos(nx) dx = 0.$$

$$B_n = \frac{1}{\pi} \int_0^{\pi} \sin(nx) dx = \frac{1 - (-1)^n}{n\pi}.$$

Hence

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \left[ \sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right].$$

## 2.10 Slide 43 - amplitude-phase conversion used in the examples

For

$$A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t) = K_n \cos(n\omega_0 t + \phi_n),$$

$$K_n = \sqrt{A_n^2 + B_n^2},$$

and the quadrant-safe phase is

$$\phi_n = \text{atan2}(-B_n, A_n).$$

## 2.11 Slide 45 - sawtooth in amplitude-phase form

For  $x(t) = t/T$  on  $0 < t < T$ , periodically extended,

$$A_0 = \frac{1}{2}, \quad A_n = 0, \quad B_n = -\frac{1}{n\pi}.$$

Therefore

$$x(t) = \frac{1}{2} - \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{\sin(n\omega_0 t)}{n}.$$

Since  $-\sin\theta = \cos(\theta + 90^\circ)$ ,

$$x(t) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \cos(n\omega_0 t + 90^\circ).$$

## 2.12 Slide 47 - quarter-duty rectangular pulse

Let  $v(t) = V_m$  for  $0 < t < T/4$  and zero for the rest of each period.

$$A_0 = \frac{V_m}{4},$$

$$A_n = \frac{V_m}{n\pi} \sin\left(\frac{n\pi}{2}\right),$$

$$B_n = \frac{V_m}{n\pi} \left[1 - \cos\left(\frac{n\pi}{2}\right)\right].$$

For the first three harmonics:

| $n$ | $A_n$         | $B_n$        | $K_n$                | $\phi_n$     |
|-----|---------------|--------------|----------------------|--------------|
| 1   | $V_m/\pi$     | $V_m/\pi$    | $\sqrt{2}V_m/\pi$    | $-45^\circ$  |
| 2   | 0             | $V_m/\pi$    | $V_m/\pi$            | $-90^\circ$  |
| 3   | $-V_m/(3\pi)$ | $V_m/(3\pi)$ | $\sqrt{2}V_m/(3\pi)$ | $-135^\circ$ |

Thus

$$v(t) = \frac{V_m}{4} + \frac{\sqrt{2}V_m}{\pi} \cos(\omega_0 t - 45^\circ) + \frac{V_m}{\pi} \cos(2\omega_0 t - 90^\circ) + \frac{\sqrt{2}V_m}{3\pi} \cos(3\omega_0 t - 135^\circ) + \dots$$

## 2.13 Slides 48-49 - trigonometric and exponential coefficients

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad C_n = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jn\omega_0 t} dt.$$

For a real signal,

$$C_0 = A_0,$$

$$C_n = \frac{A_n - jB_n}{2}, \quad C_{-n} = \frac{A_n + jB_n}{2} = C_n^*, \quad n > 0.$$

The one-sided trigonometric harmonic amplitude is

$$K_n = \sqrt{A_n^2 + B_n^2} = 2|C_n|, \quad n > 0.$$

## 2.14 Slide 51 - exponential series of the even square wave

For  $T = 2$  and  $\omega_0 = \pi$ , with  $x(t) = +1$  for  $|t| < 0.5$  and  $x(t) = -1$  for  $0.5 < |t| < 1$ ,

$$C_0 = 0,$$

and, using even symmetry,

$$C_n = \int_0^{0.5} \cos(n\pi t) dt - \int_{0.5}^1 \cos(n\pi t) dt = \frac{2\sin(n\pi/2)}{n\pi}.$$

Therefore  $C_n = 0$  for even  $n$ , while

$$C_1 = \frac{2}{\pi}, \quad C_3 = -\frac{2}{3\pi}, \quad C_5 = \frac{2}{5\pi}, \dots$$

## 2.15 Slides 53-55 - periodic rectangular pulse train

For  $f(t) = A$  over  $|t| < \tau/2$  and zero otherwise within each period  $T$ ,

$$C_0 = \frac{A\tau}{T}.$$

For  $n \neq 0$ ,

$$C_n = \frac{1}{T} \int_{-\tau/2}^{\tau/2} A e^{-jn\omega_0 t} dt.$$

$$C_n = \frac{A}{n\pi} \sin\left(\frac{n\pi\tau}{T}\right) = \frac{A\tau}{T} \operatorname{sinc}\left(\frac{n\pi\tau}{T}\right).$$

where

$$\operatorname{sinc}(u) = \frac{\sin u}{u}.$$

The coefficient values are discrete stems at  $n\omega_0$ ; the smooth sinc curve is their envelope. Increasing  $\tau$  increases the DC value and narrows the envelope.

## 2.16 Slides 56-61 - half-wave symmetry and the square-wave example

Half-wave symmetry means

$$x\left(t + \frac{T}{2}\right) = -x(t).$$

It implies

$$A_0 = 0,$$

and all even-index Fourier coefficients vanish:

$$A_n = B_n = 0 \quad \text{for even } n.$$

If the signal is also odd, only odd sine harmonics remain. If it is also even, only odd cosine harmonics remain.

For the Slide 61 square wave,  $T = 2$ ,  $\omega_0 = \pi$ , and

$$B_n = \frac{2[1 - \cos(n\pi)]}{n\pi} = \begin{cases} 4/(n\pi), & n \text{ odd,} \\ 0, & n \text{ even.} \end{cases}$$

### 3. Slide 62 - complete discussion of the nine symmetry graphs

The graphs are numbered row-wise below. Values exactly at isolated jump points do not affect Fourier coefficients, so symmetry is judged from the waveform away from those isolated points.

There are three types of symmetry:

(1) even-function symmetry

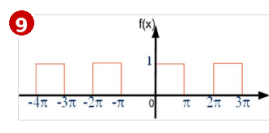
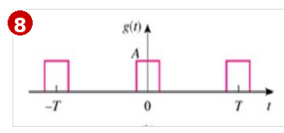
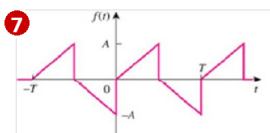
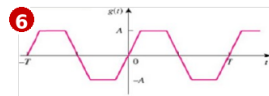
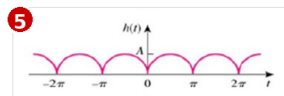
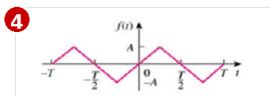
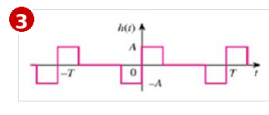
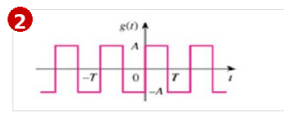
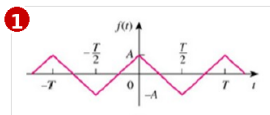
$$f(t) = f(-t)$$

(2) odd-function symmetry

$$f(t) = -f(-t)$$

(3) half-wave symmetry

$$f(t) = -f(t - T/2)$$



Numbering used in the correction note (read row-wise).

Numbered symmetry graphs from Slide 62

**Graph 1:** even and half-wave symmetric. Therefore  $A_0 = 0$ ,  $B_n = 0$ , and even-index  $A_n = 0$ : **odd cosine harmonics only**.

**Graph 2:** odd and half-wave symmetric. Therefore  $A_0 = 0$ ,  $A_n = 0$ , and even-index  $B_n = 0$ : **odd sine harmonics only**.

**Graph 3:** odd, but not half-wave symmetric. Therefore  $A_0 = A_n = 0$ : sine terms only, but both odd and even harmonic indices may occur.

**Graph 4:** odd and half-wave symmetric: odd sine harmonics only.

**Graph 5:** even, but not half-wave symmetric. Cosine terms and possibly DC may occur. The illustrated  $|\sin t|$ -type waveform has fundamental period  $\pi$ .

**Graph 6:** odd and half-wave symmetric: odd sine harmonics only.

**Graph 7:** half-wave symmetric, but neither even nor odd. Therefore  $A_0 = 0$  and all even-index coefficients vanish, but both odd cosine and odd sine terms may be required.

**Graph 8:** even, but not half-wave symmetric. Cosine terms and DC may occur; both odd and even harmonic indices may occur.

**Graph 9:** neither even, odd nor half-wave symmetric. In general  $A_0$ ,  $A_n$  and  $B_n$  may all be needed. A useful extra property is  $f(x + \pi) = 1 - f(x)$ , so  $f(x) - 1/2$  has half-wave symmetry.

#### How to test the three symmetries

- **Even:** compare points mirrored about the vertical axis:  $f(t) = f(-t)$ .
- **Odd:** compare mirrored points with sign reversal:  $f(t) = -f(-t)$ .
- **Half-wave:** shift by half of the fundamental period and check for sign reversal:  $f(t + T/2) = -f(t)$ .

A single counterexample is enough to show that a proposed symmetry is absent.

## 4. Pass-level tutorial questions

### 4.1 Slide 65, Question 1 - square wave

From the graph,

$$T = \frac{\pi}{2} \text{ s}, \quad \omega_0 = \frac{2\pi}{T} = 4 \text{ rad/s}.$$

The waveform is odd and half-wave symmetric, so  $A_0 = A_n = 0$  and  $B_n = 0$  for even  $n$ . On  $0 < t < \pi/4$ ,  $f(t) = 1$ :

$$B_n = \frac{4}{T} \int_0^{T/2} \sin(n\omega_0 t) dt.$$

$$B_n = \frac{8}{\pi} \int_0^{\pi/4} \sin(4nt) dt = \frac{2[1 - (-1)^n]}{n\pi}.$$

Hence

$$B_n = \begin{cases} 4/(n\pi), & n \text{ odd,} \\ 0, & n \text{ even.} \end{cases}$$

The corrected answer is

$$f(t) = \frac{4}{\pi} \sum_{m=0}^{\infty} \frac{\sin(4(2m+1)t)}{2m+1}.$$

The printed sum over every positive integer  $n$  is incorrect because it introduces even harmonics.

### 4.2 Slide 65, Question 2 - odd trapezoidal waveform

From the figure,  $T = \pi$  and  $\omega_0 = 2$ . Over  $0 \leq t \leq \pi/2$ ,

$$f(t) = \begin{cases} -12t/\pi, & 0 \leq t \leq \pi/6, \\ -2, & \pi/6 \leq t \leq \pi/3, \\ 12t/\pi - 6, & \pi/3 \leq t \leq \pi/2. \end{cases}$$

The waveform is odd and half-wave symmetric, so only odd-index sine coefficients can remain. Use

$$B_n = \frac{4}{\pi} \int_0^{\pi/2} f(t) \sin(2nt) dt.$$

Substituting the three pieces,

$$B_n = \frac{4}{\pi} \left[ \int_0^{\pi/6} \left(-\frac{12t}{\pi}\right) \sin(2nt) dt + \int_{\pi/6}^{\pi/3} (-2) \sin(2nt) dt + \int_{\pi/3}^{\pi/2} \left(\frac{12t}{\pi} - 6\right) \sin(2nt) dt \right].$$

After integration and simplification,

$$B_n = -\frac{12}{\pi^2 n^2} \left[ \sin\left(\frac{n\pi}{3}\right) + \sin\left(\frac{2n\pi}{3}\right) \right].$$

For odd  $n$ ,  $\sin(2n\pi/3) = \sin(n\pi/3)$ , while the even coefficients vanish by half-wave symmetry. Therefore

$$f(t) = -\frac{24}{\pi^2} \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{\sin(n\pi/3)}{n^2} \sin(2nt).$$

The restriction  $n$  odd is required in the compact answer shown on the slide.

### 4.3 Slide 65, Question 3 - half-wave-rectified sine

Let

$$f(t) = \begin{cases} \sin(\pi t), & 0 < t < 1, \\ 0, & 1 < t < 2, \end{cases} \quad T = 2, \quad \omega_0 = \pi.$$

The average is

$$A_0 = \frac{1}{2} \int_0^1 \sin(\pi t) dt = \frac{1}{\pi}.$$

For the sine coefficient,

$$B_1 = \int_0^1 \sin^2(\pi t) dt = \frac{1}{2},$$

and orthogonality gives  $B_n = 0$  for  $n \neq 1$ .

For  $n \neq 1$ ,

$$A_n = \int_0^1 \sin(\pi t) \cos(n\pi t) dt.$$

Using

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)],$$

we obtain

$$A_n = \frac{1 - \cos((n+1)\pi)}{\pi(1 - n^2)}.$$

Thus

$$f(t) = \frac{1}{\pi} + \frac{1}{2} \sin(\pi t) + \sum_{n=2}^{\infty} \frac{1 - \cos((n+1)\pi)}{\pi(1 - n^2)} \cos(n\pi t).$$

For the fourth harmonic,

$$A_4 = \frac{1 - \cos(5\pi)}{\pi(1 - 16)} = -\frac{2}{15\pi}.$$

Therefore the fourth-harmonic term is

$$-\frac{2}{15\pi} \cos(4\pi t).$$

### 4.4 Slide 65, Question 4 - symmetric sawtooth

For

$$f(t) = \frac{2t}{T}, \quad -\frac{T}{2} < t < \frac{T}{2},$$

periodically extended, the waveform is odd. Therefore  $A_0 = A_n = 0$  and

$$B_n = \frac{4}{T} \int_0^{T/2} \frac{2t}{T} \sin(n\omega_0 t) dt.$$

Integration by parts gives

$$B_n = \frac{2(-1)^{n-1}}{n\pi}.$$

Hence

$$f(t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin(n\omega_0 t).$$

#### 4.5 Slide 68, Question 5 - symmetry classification

##### Part (a)

The period is 8. It is not even because, for example,  $v(1) = 5$  but  $v(-1) = 0$ . It is not odd because  $v(-1) \neq -v(1)$ . It has half-wave symmetry because  $v(t + 4) = -v(t)$ .

**Answer:** not even; not odd; half-wave symmetric;  $T = 8$ .

##### Part (b)

The period is 8. At  $t = 2$ ,  $v(2) = 10$ , but  $v(6) = 0$ , so half-wave symmetry fails. Simple mirrored-point checks also disprove even and odd symmetry.

**Answer:** no even, odd or half-wave symmetry;  $T = 8$ .

##### Part (c)

The period is 4. The periodic extension satisfies  $v(-t) = -v(t)$ , so it is odd. It is not even and does not satisfy half-wave symmetry.

**Answer:** odd only;  $T = 4$ .

##### Part (d) - Fourier series of waveform (a)

For part (a),  $T = 8$  and  $\omega_0 = \pi/4$ . Half-wave symmetry gives  $A_0 = 0$  and eliminates all even harmonics.

For odd  $n$ ,

$$A_n = \frac{1}{2} \int_0^2 5 \cos\left(\frac{n\pi t}{4}\right) dt = \frac{10}{n\pi} \sin\left(\frac{n\pi}{2}\right),$$

$$B_n = \frac{1}{2} \int_0^2 5 \sin\left(\frac{n\pi t}{4}\right) dt = \frac{10}{n\pi}.$$

Therefore

$$v(t) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{10}{n\pi} \left[ \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi t}{4}\right) + \sin\left(\frac{n\pi t}{4}\right) \right].$$

#### 4.6 Slide 68, Question 6 - complex Fourier coefficient

The waveform has  $T = 2$ ,  $\omega_0 = \pi$ , and

$$v(t) = \begin{cases} 1, & 0 < t < 1, \\ -1, & 1 < t < 2. \end{cases}$$

For  $n \neq 0$ ,

$$C_n = \frac{1}{2} \left[ \int_0^1 e^{-jn\pi t} dt - \int_1^2 e^{-jn\pi t} dt \right].$$

$$C_n = \frac{1 - (-1)^n}{jn\pi} = -j \frac{1 - (-1)^n}{n\pi}.$$

Therefore

$$C_n = \begin{cases} -\frac{j2}{n\pi}, & n \text{ odd,} \\ 0, & n \text{ even,} \end{cases} \quad C_0 = 0.$$

The answer box should be labelled  $C_n$ , not  $v(t)$ .

#### 4.7 Slide 68, Question 7 - reconstructing from an amplitude spectrum

The lines occur at 500, 1000 and 1500 Hz, so

$$f_0 = 500 \text{ Hz}, \quad \omega_0 = 2\pi f_0 = 1000\pi \text{ rad/s.}$$

Odd symmetry gives  $A_0 = A_n = 0$ . Because all coefficients are stated to be nonnegative, the line heights are

$$B_1 = 0.2, \quad B_2 = 0.6, \quad B_3 = 0.4.$$

Thus

$$f(t) = 0.2\sin(1000\pi t) + 0.6\sin(2000\pi t) + 0.4\sin(3000\pi t).$$

## 5. Selected higher-level items requiring correction or fuller working

### 5.1 Slide 71, Question 1 - correcting and deriving the coefficient formulas

The general formulas must integrate over a complete period. Start with

$$x(t) = A_0 + \sum_{n=1}^{\infty} [A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t)].$$

Multiply both sides by  $\cos(m\omega_0 t)$  and integrate over one complete period. Orthogonality makes every term vanish except the  $m$ th cosine term:

$$\int_T x(t) \cos(m\omega_0 t) dt = A_m \int_T \cos^2(m\omega_0 t) dt = A_m \frac{T}{2}.$$

Therefore

$$A_m = \frac{2}{T} \int_T x(t) \cos(m\omega_0 t) dt.$$

Similarly,

$$B_m = \frac{2}{T} \int_T x(t) \sin(m\omega_0 t) dt.$$

A convenient explicit interval is  $[-T/2, T/2]$  or  $[t_0, t_0 + T]$ . The interval  $[0, T/2]$  is valid only after invoking a suitable symmetry and changing the factor accordingly.

### 5.2 Slide 71, Question 3 - spectra before and after rectification

The source is

$$v(t) = 120\sqrt{2} \sin(2\pi 60t) \text{ V},$$

so

$$V_m = 120\sqrt{2} \approx 169.7 \text{ V}.$$

#### Original sine

A one-sided trigonometric spectrum has one line of amplitude 169.7 V at 60 Hz and no DC.

#### Half-wave rectified

$$V_{DC} = \frac{V_m}{\pi} \approx 54.02 \text{ V},$$

$$60\text{-Hz sine amplitude} = \frac{V_m}{2} \approx 84.85 \text{ V}.$$

The cosine coefficients at  $2m(60)$  Hz have magnitudes

$$\frac{2V_m}{\pi(4m^2 - 1)}.$$

The first few magnitudes are:

- 120 Hz: 36.01 V
- 240 Hz: 7.20 V
- 360 Hz: 3.09 V

### Full-wave rectified

$$V_{\text{DC}} = \frac{2V_m}{\pi} \approx 108.04 \text{ V.}$$

There is no 60-Hz component. The cosine coefficients at  $2m(60)$  Hz have magnitudes

$$\frac{4V_m}{\pi(4m^2 - 1)}.$$

The first few magnitudes are:

- 120 Hz: 72.03 V
- 240 Hz: 14.41 V
- 360 Hz: 6.17 V
- 480 Hz: 3.43 V

The signs of the cosine coefficients indicate phase; an amplitude spectrum shows their nonnegative magnitudes.

## 6. Use of this note

Use this note together with the Week 1 lecture slides. In an assessment, show the setup, symmetry argument, integration interval and coefficient calculation rather than writing only the final series.